

# BIOLOGY 30

## Unit B — Reproduction and Development Study Guide

### 1 Unit / Module Objectives

| Topic Area                    | Diploma-Level Objective                                                                                                                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Male Reproductive System      | Analyze structures of the male reproductive system and explain how testes, seminiferous tubules, Sertoli cells and interstitial cells support sperm production, maturation and hormonal regulation. |
| Female Reproductive System    | Analyze structures of the female reproductive system and explain how ovaries, oviducts, uterus and endometrium support ovulation, fertilization, implantation and pregnancy.                        |
| Hormonal Regulation           | Explain how GnRH, FSH, LH, testosterone, estrogen and progesterone regulate reproductive function through negative and positive feedback.                                                           |
| Cycle and Data Analysis       | Interpret hormone graphs and experimental reproductive data to predict ovulation, menstruation, infertility or successful implantation.                                                             |
| Fertilization and Development | Explain acrosome reaction, prevention of polyspermy, cleavage, blastocyst formation, implantation and early embryonic support.                                                                      |
| Technology and STS            | Evaluate reproductive technologies, fertility treatments and developmental risk factors using biological evidence.                                                                                  |

### 2 Student Self-Verification Checklist

| Topic       | Self-Verification Statement                                                                                                                              |
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| Male System | <input type="checkbox"/> I can justify how each male reproductive structure contributes to sperm production, maturation, transport or hormone secretion. |

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| Female System               | <input type="checkbox"/> I can justify how each female reproductive structure contributes to ovulation, fertilization, implantation or pregnancy. |
| Hormones                    | <input type="checkbox"/> I can explain negative and positive feedback involving GnRH, FSH, LH, testosterone, estrogen and progesterone.           |
| Graph Analysis              | <input type="checkbox"/> I can interpret hormone graphs and predict ovulation, menstruation, pregnancy or infertility from the data.              |
| Fertilization / Development | <input type="checkbox"/> I can explain acrosome reaction, prevention of polyspermy, cleavage, blastocyst formation and implantation.              |
| Technology / Evidence       | <input type="checkbox"/> I can analyze fertility experiment data and evaluate reproductive technologies or developmental risk factors.            |

### 3 Diploma-Level Practice Questions with Fully Worked Solutions

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| <b>Question</b>                     | Hormone graph interpretation: A 28-day cycle graph shows estrogen rising from days 5–12, a sharp LH spike on day 13, progesterone rising from days 15–24, and both estrogen and progesterone falling on days 25–28. Identify when ovulation occurs and explain the hormonal control of the cycle. |
| <b>Step 1 — Identification</b>      | Relevant variables are estrogen, LH and progesterone across the menstrual cycle. Ovulation is associated with the LH surge, while progesterone reflects corpus luteum activity after ovulation.                                                                                                   |
| <b>Step 2 — Principle Statement</b> | Sustained high estrogen from the mature follicle causes positive feedback on the hypothalamus and anterior pituitary, triggering the LH surge. After ovulation, the corpus luteum secretes progesterone, which maintains the endometrium. Falling progesterone and estrogen cause menstruation.   |

**Step 3 — Explicit Reasoning**

The LH spike on day 13 indicates ovulation will occur about 24–36 hours later, so ovulation occurs around day 14. Before this, rising estrogen from the dominant follicle stimulates follicle maturation. Once estrogen remains high, positive feedback triggers the LH surge. After ovulation, the ruptured follicle becomes the corpus luteum, raising progesterone from days 15–24. Progesterone maintains the endometrium. When the corpus luteum degenerates late in the cycle, progesterone and estrogen fall, the uterine lining loses support and menstruation begins.

**Step 4 — Validation**

This matches the normal sequence follicular phase → LH surge → ovulation → luteal phase → menstruation.

**Step 5 — Final Answer**

Ovulation occurs around day 14. Rising estrogen triggers the LH surge, the corpus luteum raises progesterone after ovulation, and the late drop in progesterone and estrogen causes menstruation.

**Common Error**

Students often place ovulation at the progesterone peak instead of just after the LH surge.

**Exam Strategy**

On cycle graphs, find the LH spike first. That anchors ovulation and helps explain the rest of the pattern.

**Question**

Male fertility experiment: In a study, one group receives an LH receptor blocker. Compared with the control group, testosterone concentration falls sharply, seminiferous tubules become less active, and sperm count declines after several weeks. Explain the pathway linking the treatment to the reduced sperm count.

**Step 1 — Identification**

Relevant structures are the anterior pituitary, LH receptors on interstitial cells, testosterone and the seminiferous tubules where spermatogenesis occurs.

**Step 2 — Principle Statement**

LH stimulates interstitial cells to secrete testosterone. Testosterone is required, along with Sertoli cell support, for normal spermatogenesis. If LH signaling is blocked, testosterone falls and sperm production decreases.

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| <b>Step 3 — Explicit Reasoning</b>  | The LH receptor blocker prevents interstitial cells from responding to LH. As a result, testosterone secretion drops. Testosterone normally supports the seminiferous tubules and helps maintain the environment needed for meiosis and sperm differentiation. When testosterone remains low for several weeks, spermatogenesis slows because developing sperm no longer receive the hormonal support required for continued maturation. That is why seminiferous tubules become less active and sperm count eventually declines. |
| <b>Step 4 — Validation</b>          | The delayed drop in sperm count is biologically reasonable because sperm production takes time and does not stop instantly.                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Step 5 — Final Answer</b>        | Blocking LH receptors lowers testosterone production by interstitial cells, which reduces support for spermatogenesis in the seminiferous tubules and causes sperm count to fall.                                                                                                                                                                                                                                                                                                                                                 |
| <b>Common Error</b>                 | Students often say LH acts directly on sperm cells; it acts mainly on interstitial cells.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Exam Strategy</b>                | Separate hormone targets first: LH → interstitial cells, FSH → Sertoli cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Question</b>                     | A woman is given a GnRH inhibitor beginning on day 3 of her cycle. Predict the effects on FSH, LH, follicle development, estrogen concentration and ovulation.                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Step 1 — Identification</b>      | The relevant endocrine axis is hypothalamus → GnRH → anterior pituitary → FSH/LH → ovary.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Step 2 — Principle Statement</b> | GnRH stimulates pituitary release of FSH and LH. If GnRH is inhibited early in the cycle, follicle maturation is impaired, estrogen stays low and the LH surge does not occur.                                                                                                                                                                                                                                                                                                                                                    |
| <b>Step 3 — Explicit Reasoning</b>  | Because the inhibitor blocks GnRH action from day 3, the anterior pituitary releases less FSH and LH. Reduced FSH slows development of ovarian follicles, so the dominant follicle does not mature properly. Since the follicle remains underdeveloped, estrogen secretion does not rise to the high level needed for positive feedback. Without that estrogen peak, there is no LH surge. If there is no LH surge, ovulation does not occur and a normal corpus luteum does not form.                                            |
| <b>Step 4 — Validation</b>          | This prediction is consistent because the ovarian cycle depends on pituitary gonadotropins from the start of the follicular phase.                                                                                                                                                                                                                                                                                                                                                                                                |

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| <b>Step 5 — Final Answer</b>        | FSH and LH decrease, follicle development is impaired, estrogen remains too low for positive feedback, and ovulation is prevented.                                                                                                                                                                                                                                                               |
| <b>Common Error</b>                 | Students often predict only LH changes, but both FSH and LH depend on GnRH stimulation.                                                                                                                                                                                                                                                                                                          |
| <b>Exam Strategy</b>                | Move stepwise down the endocrine axis and predict the consequence at each level.                                                                                                                                                                                                                                                                                                                 |
| <b>Question</b>                     | Fertilization data analysis: In an in vitro fertilization trial, egg cells are exposed to sperm with a damaged acrosome. Most sperm reach the zona pellucida, but fertilization rate remains extremely low. Explain why the fertilization rate stays low even though sperm contact the egg.                                                                                                      |
| <b>Step 1 — Identification</b>      | Relevant structures are the acrosome, zona pellucida, sperm membrane and egg plasma membrane.                                                                                                                                                                                                                                                                                                    |
| <b>Step 2 — Principle Statement</b> | The acrosome contains enzymes required to digest the outer layers surrounding the egg. Contact alone is insufficient if the sperm cannot penetrate the zona pellucida.                                                                                                                                                                                                                           |
| <b>Step 3 — Explicit Reasoning</b>  | Sperm with a damaged acrosome may swim normally and still reach the egg surface. However, without normal acrosomal enzymes, they cannot digest a path through the zona pellucida. If they cannot cross this layer, they cannot fuse with the egg membrane and cannot deliver their nucleus into the egg. Therefore, fertilization remains low even though many sperm physically contact the egg. |
| <b>Step 4 — Validation</b>          | This result is biologically consistent because the acrosome reaction is an essential step before membrane fusion.                                                                                                                                                                                                                                                                                |
| <b>Step 5 — Final Answer</b>        | Fertilization stays low because damaged acrosomes prevent sperm from penetrating the zona pellucida, so membrane fusion cannot occur.                                                                                                                                                                                                                                                            |
| <b>Common Error</b>                 | Students often think the acrosome helps only with sperm motility rather than egg penetration.                                                                                                                                                                                                                                                                                                    |
| <b>Exam Strategy</b>                | Ask which structure solves which barrier: motility reaches the egg, acrosome enzymes penetrate it.                                                                                                                                                                                                                                                                                               |
| <b>Question</b>                     | Explain how birth control pills prevent pregnancy, and predict what would happen to ovulation if the pills were stopped halfway through the cycle.                                                                                                                                                                                                                                               |
| <b>Step 1 — Identification</b>      | Relevant factors are synthetic estrogen and progesterone, FSH, LH, ovulation and the endometrium.                                                                                                                                                                                                                                                                                                |

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| <b>Step 2 — Principle Statement</b> | Birth control pills maintain hormone levels that suppress FSH and LH through negative feedback, preventing the normal follicular peak and LH surge.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Step 3 — Explicit Reasoning</b>  | While the pills are taken, synthetic ovarian hormones signal that hormone levels are already sufficient. The hypothalamus releases less GnRH, and the pituitary releases less FSH and LH. Lower FSH limits follicle maturation and the absence of a strong LH surge prevents ovulation. If the pills are stopped halfway through the cycle, suppression is removed. Gonadotropin secretion may begin to recover, allowing a follicle to continue maturing. Depending on timing, estrogen could eventually rise enough to trigger an LH surge, so ovulation may become possible again. |
| <b>Step 4 — Validation</b>          | This is consistent because the pill mainly prevents ovulation by suppressing the endocrine pattern needed for it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Step 5 — Final Answer</b>        | Birth control pills prevent pregnancy by suppressing FSH and LH through negative feedback. If they are stopped mid-cycle, hormonal suppression can lift and ovulation may occur if follicle maturation resumes.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Common Error</b>                 | Students often say the pill kills sperm or always blocks implantation directly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Exam Strategy</b>                | State the primary mechanism first: no normal LH surge means no ovulation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Question</b>                     | Development sequence: Describe the events from fertilization to implantation, and explain why cleavage increases cell number without increasing embryo size.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Step 1 — Identification</b>      | Relevant stages are fertilization, zygote, cleavage, morula, blastocyst and implantation in the endometrium.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Step 2 — Principle Statement</b> | After fertilization in the oviduct, the diploid zygote undergoes rapid mitotic divisions called cleavage. These divisions partition the original cytoplasm into many smaller cells while the embryo moves toward the uterus.                                                                                                                                                                                                                                                                                                                                                          |

**Step 3 — Explicit Reasoning**

Fertilization forms a diploid zygote. As it travels toward the uterus, the zygote divides repeatedly by mitosis to produce many smaller blastomeres. Because these divisions occur without significant growth between them, the embryo contains more cells but remains about the same overall size. Continued cleavage forms a morula and then a blastocyst with an inner cell mass and trophoblast. Once the blastocyst reaches the uterus, the trophoblast begins embedding into the endometrium, initiating implantation and early placenta formation.

**Step 4 — Validation**

This sequence is correct because implantation requires a blastocyst, not a single-cell zygote, and cleavage precedes implantation.

**Step 5 — Final Answer**

Fertilization forms a zygote, cleavage produces many smaller cells without increasing total size, a blastocyst forms, and the blastocyst implants in the uterine lining.

**Common Error**

Students often confuse cleavage with growth or place implantation before blastocyst formation.

**Exam Strategy**

Memorize the order: fertilization → zygote → cleavage → morula → blastocyst → implantation.

**Question**

Pregnancy hormone interpretation: A patient has a positive pregnancy test, but progesterone levels begin dropping during week 4. Explain why this pattern is dangerous and identify the hormone interaction that should normally prevent it.

**Step 1 — Identification**

Relevant hormones are hCG and progesterone; relevant structures are the implanted embryo, corpus luteum and endometrium.

**Step 2 — Principle Statement**

After implantation, embryonic tissues produce hCG. hCG maintains the corpus luteum so progesterone stays high. Progesterone is required to keep the endometrium intact.

**Step 3 — Explicit Reasoning**

A positive pregnancy test indicates hCG is present. Normally, that hCG rescues the corpus luteum from degeneration. If progesterone is still dropping in week 4, the corpus luteum may not be responding adequately or hCG support may be insufficient. Falling progesterone causes the endometrium to lose support, increasing the risk that the lining will begin to break down. That threatens the implanted embryo because early pregnancy depends on a stable uterine lining until the placenta can take over hormone production.

**Step 4 — Validation**

This is consistent because early pregnancy cannot be maintained if progesterone falls too soon.

**Step 5 — Final Answer**

The pattern is dangerous because low progesterone can destabilize the endometrium and threaten pregnancy. Normally, hCG maintains the corpus luteum so progesterone remains high.

**Common Error**

Students often say hCG directly maintains the embryo rather than indirectly maintaining pregnancy through the corpus luteum.

**Exam Strategy**

Link hormones to targets: hCG → corpus luteum; progesterone → endometrium.

**Question**

Compare spermatogenesis and oogenesis in terms of timing of meiosis, cytoplasmic division and final products, and explain why the ovum is much larger than a sperm cell.

**Step 1 — Identification**

Both processes use meiosis to produce haploid gametes, but they differ in timing, distribution of cytoplasm and number of functional products.

**Step 2 — Principle Statement**

Spermatogenesis is continuous after puberty and produces four functional sperm. Oogenesis begins before birth, pauses for long periods and produces one large ovum plus polar bodies through unequal division.

**Step 3 — Explicit Reasoning**

In spermatogenesis, the goal is to produce many small motile cells, so meiosis produces four sperm with minimal cytoplasm. In oogenesis, most cytoplasm is retained in a single cell so the ovum contains nutrient reserves, organelles and molecular resources needed for early development after fertilization. This is why meiosis in females produces one large ovum and polar bodies rather than four equal gametes. Timing also differs: oocytes form before birth and may remain arrested for years, whereas sperm production proceeds continuously after puberty.

**Step 4 — Validation**

This comparison is valid because sperm are specialized for delivery of genetic material, while ova are specialized for supporting early development.

**Step 5 — Final Answer**

Spermatogenesis is continuous and yields four small sperm, whereas oogenesis is discontinuous and yields one large ovum. The ovum is larger because it retains most of the cytoplasm needed for early embryonic development.

**Common Error**

Students often say the ovum is larger only because it is female rather than because of its developmental role.



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| <b>Exam Strategy</b>                | Compare function first, then structure: mobility favors small sperm, support favors a large ovum.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Question</b>                     | Teratogen data analysis: In a study of prenatal alcohol exposure, embryos exposed during early organ formation show a much higher rate of structural abnormalities than embryos exposed later in pregnancy. Explain why early exposure is more damaging.                                                                                                                                                                                                                                                                                                                      |
| <b>Step 1 — Identification</b>      | Relevant concepts are embryonic development, organogenesis, cell differentiation and teratogens.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Step 2 — Principle Statement</b> | Teratogens are most harmful when major organs and body plans are being established because disruption at that stage affects many downstream developmental pathways.                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Step 3 — Explicit Reasoning</b>  | During early embryonic development, cells are rapidly dividing, differentiating and organizing into tissues and organ systems. If alcohol disrupts signaling, cell migration or gene expression during this stage, the structure of organs can be altered before they are fully established. Later exposure can still cause damage, but many basic body structures are already formed. Therefore, early exposure tends to produce more severe structural abnormalities because it interferes with organ formation at the time when developmental pathways are most sensitive. |
| <b>Step 4 — Validation</b>          | This pattern is consistent with the principle that timing of exposure is critical in developmental biology.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Step 5 — Final Answer</b>        | Early exposure is more damaging because teratogens disrupt organ formation during the most sensitive stage of embryonic development.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Common Error</b>                 | Students often think the amount of exposure is the only factor and ignore timing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Exam Strategy</b>                | When interpreting developmental data, ask not only what the teratogen is, but when exposure occurred.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Question</b>                     | Evaluate one biological benefit and one biological limitation of in vitro fertilization (IVF) using reproductive physiology.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Step 1 — Identification</b>      | Relevant factors include ovulation, fertilization outside the body, embryo transfer and implantation success.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Step 2 — Principle Statement</b> | IVF can bypass barriers to natural fertilization, but successful implantation and pregnancy still depend on normal embryonic development and uterine conditions.                                                                                                                                                                                                                                                                                                                                                                                                              |

**Step 3 — Explicit Reasoning**

A major biological benefit of IVF is that fertilization can occur even when natural meeting of sperm and egg is unlikely, such as when oviducts are blocked or sperm numbers are low. Fertilization is achieved in a controlled environment and an embryo can then be transferred to the uterus. A biological limitation is that fertilization alone does not guarantee pregnancy. The embryo must still implant successfully, and the uterus must provide hormonal and structural support for development. Therefore, implantation failure remains a major limitation even when fertilization is achieved.

**Step 4 — Validation**

This evaluation is balanced because it distinguishes success at fertilization from success at implantation and pregnancy.

**Step 5 — Final Answer**

IVF can bypass barriers to natural fertilization, but pregnancy may still fail if the embryo does not implant or the uterine environment is unsuitable.

**Common Error**

Students often assume IVF guarantees pregnancy once an embryo is produced.

**Exam Strategy**

Separate the stages: ovulation, fertilization, implantation and pregnancy maintenance are different biological hurdles.